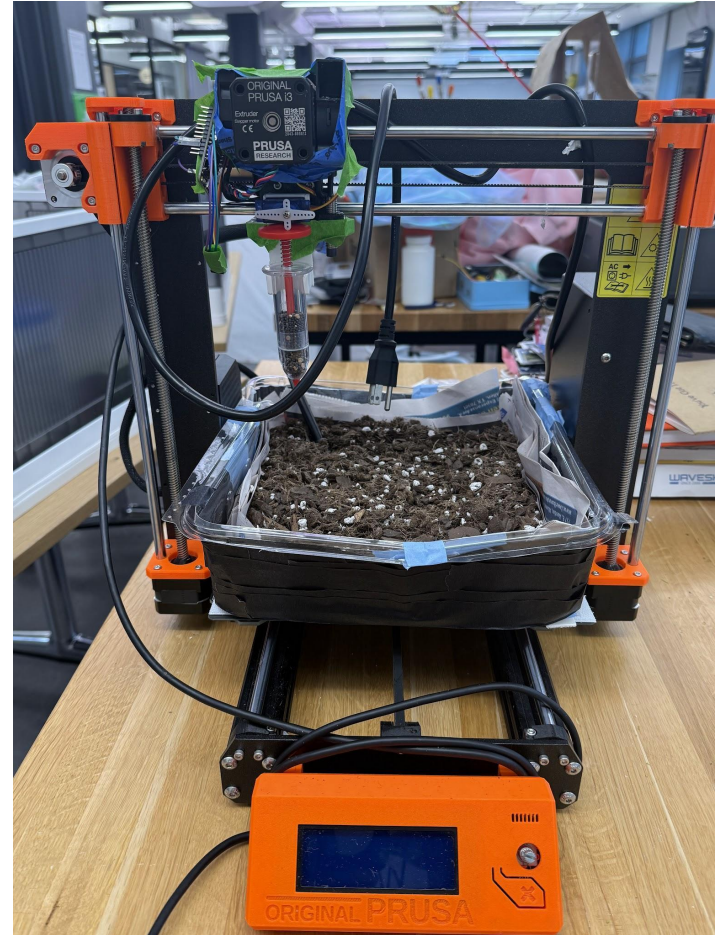


Seed Dropper System: Practicing Resilience Through Reuse

-Abdul Q

Project Overview

- Built a **seed dropper system**
- Repurposed a **broken 3D printer**
- Controlled via a **custom GUI app**
- Drops seeds precisely onto soil



Why This Project

- Focus on **care + management**
- Reuse instead of discard
- Small-scale **home farming support**
- Systems thinking through hardware + software

System Breakdown

Physical System

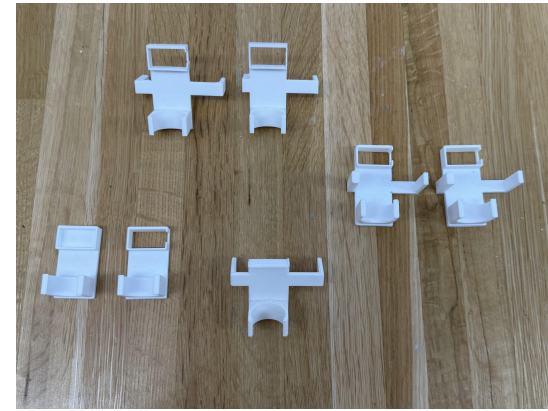
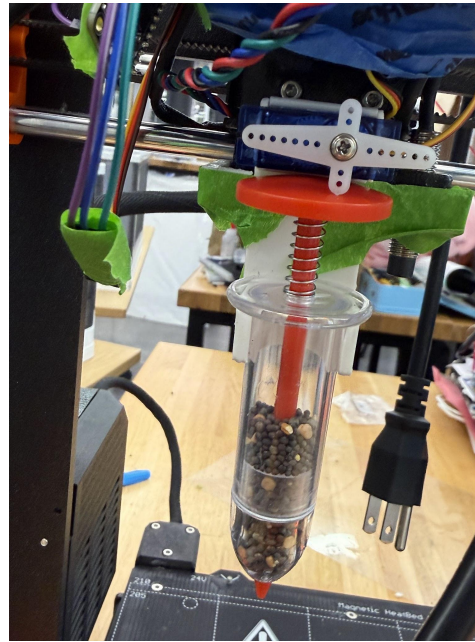
- Seed tube
- 3D printed holder
- Servo motor

Control System

- ESP32 microcontroller
- Python GUI (Mac)

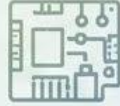
Movement System

- Pronterface → controls printer head



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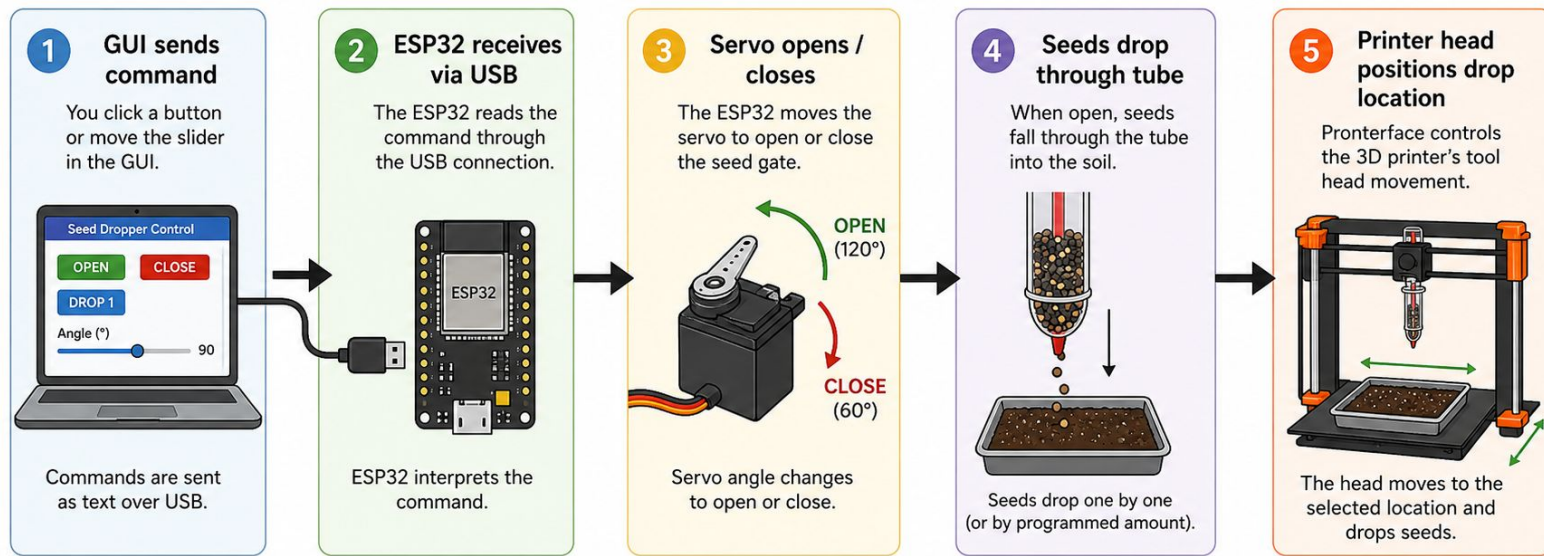
Movement System

Pronterface →
controls printer
head



How It Works

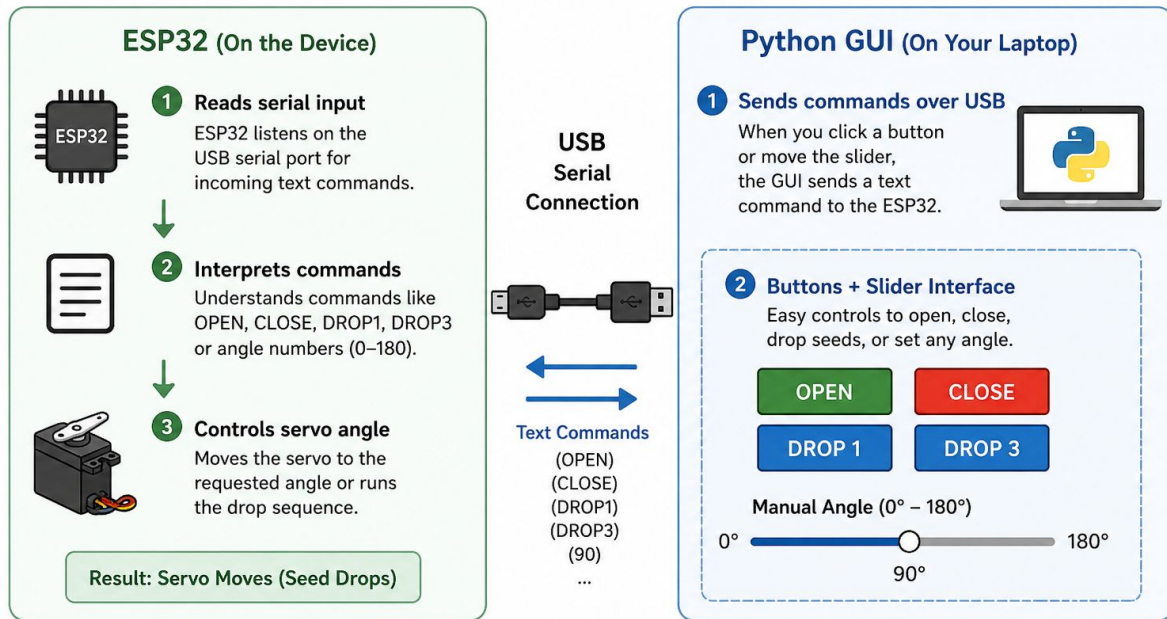
How the Seed Dropper System Works



From Click to Seed: Software command → USB → ESP32 → Servo → Seed Drop → Precise Placement



Code Summary



KEY IDEA:



Text Commands



Hardware Movement
(Servo Angle Changes)

Simple text over USB makes real-world things move.



What Forms of Care Became Visible?

- Care for **materials** (reuse)
- Care for **food systems**
- Care through **precision + efficiency**
- Care as **maintenance and adaptation**

Limits of the Project

Hardware Limits

- Power system not finalized
- Wired communication only
- Servo precision limitations

Physical Limits

- Soil containment design weak
- Seed flow inconsistency

Design:

- Two columns: Hardware / Physical

Challenges

- Time constraints (senior project overlap)
- 3D modeling the holder
- Debugging hardware + software together
- Physical iteration vs digital design

How Could This Continue?

- Wireless control system
- Custom PCB
- Integrated enclosure
- Soil + moisture sensing
- Automated watering system

Reflection

- Independent work → full control + responsibility
- Revealed limits of working alone
- Required balancing design + engineering
- Showed value of small, practical systems

One Take Away

Resilience is not building new systems, but adapting and sustaining existing ones.